

Towards a Better Understanding of Adam and Muon

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Goal of This Talk

Goal: Principled foundation behind **Adam & Muon** and how to improve

$$\min_{\theta} \ell(\theta) := \frac{1}{n} \sum_{i=1}^n \ell_i(\theta)$$

- sample size or # mini-batches: n
- NN parameters: $\theta \in \mathbb{R}^d$
- loss on the i -th batch: $\ell_i(\theta)$

- Adam [Kingma and Ba 15]:

Given a flattened gradient vector $g \in \mathbb{R}^d$:

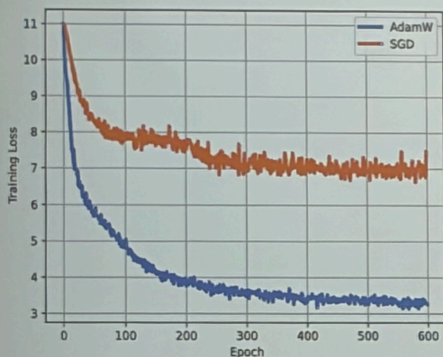
- $m_k = (1 - \beta_1)g + \beta_1 m_{k-1}$
- $v_k = (1 - \beta_2)g \circ g + \beta_2 v_{k-1}$
- $\theta_{k+1} = \theta_k - \eta_k m_k / \sqrt{v_k}$

- Muon [Jordan et al. 24]:

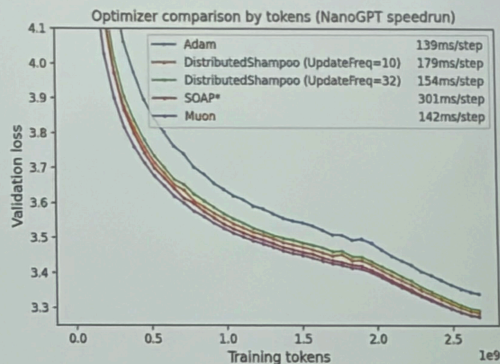
Given a 2D hidden-layer gradient matrix $G \in \mathbb{R}^{m \times n}$:

- $M_k = (1 - \beta)G + \beta M_{k-1}$
- $W_{k+1} = W_k - \eta_k NS(M_k)$
- $NS(\cdot)$: Newton-Schultz iteration $\approx (M_k M_k^T)^{-\frac{1}{2}} M_k$

SGD < AdamW < Muon for LLMs



GPT2-125M: SGD < AdamW



GPT2-125M: AdamW < Muon

- **Muon is now used in:** Kimi K2, Zhipu GLM 5.1, DeepSeek V4, OpenAI (rumor)
- **SGD < AdamW < Muon, but why? How to improve?**

Overview of This Talk

Part I: Safety and guarantees

- Does Adam diverge or converge?
- How to choose hyperparams to avoid divergence?
- Divergence & Convergence of Muon?

Part II: Special Problem structure

- $\text{SGD} < \text{Adam} < \text{Muon}$ on LLMs
- What problem structure drives this gap?

Part III : How to improve?

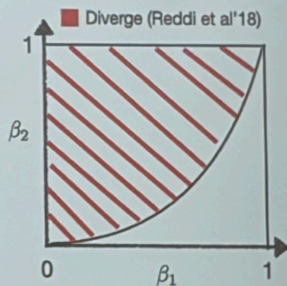
- Can we exploit NN-specific structure to improve Adam and Muon?

Negative Results on Adam

- Despite popularity, Adam is claimed to **diverge** even on simple convex function
- Reddi et al.' 18 (ICLR 2018 Best Paper):

[Kingma and Ba 15]:
provided a convergence proof
for any $\beta_1 < \sqrt{\beta_2}$, $0 < \beta_1, \beta_2 < 1$

[Reddi et al. 18]:
Identify **an error** in the proof.
Claim divergence in the same region



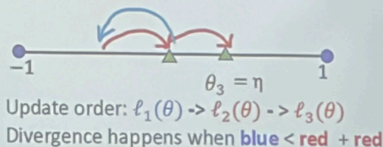
[Reddi et al. 18]: For any β_1, β_2 s.t. $\beta_1 < \sqrt{\beta_2}$, there exists a problem s.t. Adam diverges

Reddi, S.J., Kale, S. and Kumar, S., On the convergence of adam and beyond, ICLR 2018, Best Paper Award

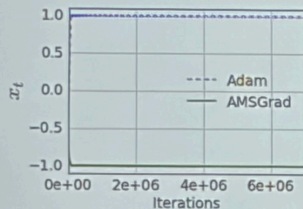
Why Diverge? [Reddi et al. 2018]

$$\min \sum_{i=1}^n \ell_i(\theta), \text{ s.t., } \theta \in [-1, 1], n \geq 3$$

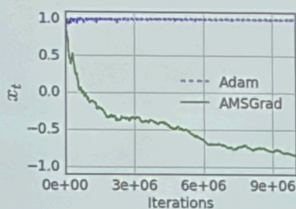
$$\ell_i(\theta) = \begin{cases} n\theta & \text{if } i = 1 \\ -\theta & \text{otherwise} \end{cases}$$



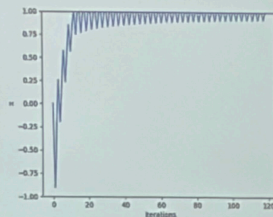
- global optimum: $\theta^* = -1$
- Does **randomized update order** help prevent divergence? **No**. Proved in Thm3 Reddi et al.
- Does **diminishing stepsize** help prevent divergence? **No**. $\eta_k \propto \frac{1}{\sqrt{k}}$ is already used in the proof



(a) Cyclic update order



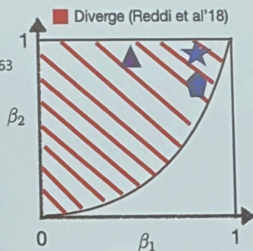
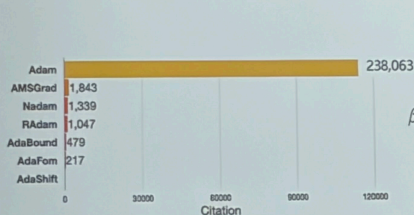
(b) Randomized update order



(c) Diminishing stepsize

Inconsistence with Practice

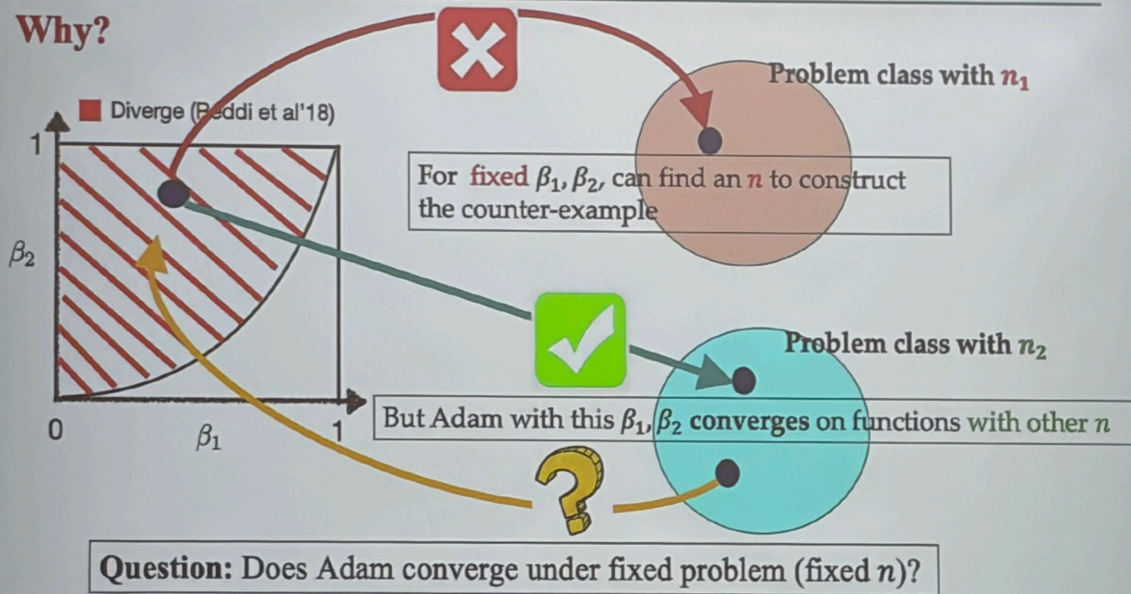
- Many new variants are proposed: AMSGrad, AdaBound, AdaFom, AdaShift, Adaptive β_{1k} & β_{2k} , ...
[Reddi et al.18, Luo et al.19, Chen et al.18, Zou et al. 19, Zhou et al. 18, Chen et al. 22, ...]
- Meanwhile, **inconsistent** with practice: Adam is still dominant



- ★ Most deep learning tasks
(e.g. RL, NLP, CV, GAN, etc.):
 $\beta_1 = 0.9, \beta_2 = 0.999$
- ▲ Conditional GAN, DCGAN, etc:
 $\beta_1 = 0.5, \beta_2 = 0.999$
- ⬠ LLMs
 $\beta_1 = 0.9, \beta_2 = 0.95$

Why is the divergence not observed?

Why?



Why?

Proof (Reddi et al. 18):

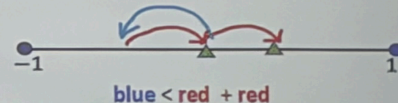
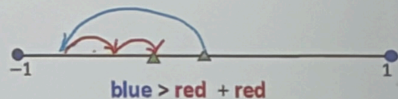
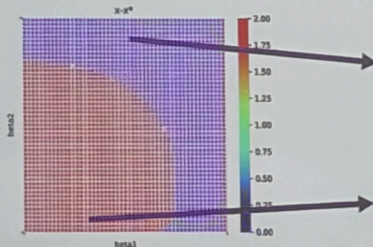
For any **fixed** β_1, β_2 s.t. $\beta_1 < \sqrt{\beta_2}$, we can find an n to construct the divergence example $\ell(\theta)$

- **An important feature:** Reddi et al. fix β_1, β_2 **before** picking the problem (change n according to β_1, β_2)
- **While in practice:** hyperparameters are often problem-dependent, e.g. tuning $\text{lr} = 1/L$ for GD

Question: Does Adam converge under fixed problem class (fixed n)?

$$\ell_i(\theta) = \begin{cases} n\theta & \text{if } i = 1 \\ -\theta & \text{otherwise} \end{cases}$$

$$\theta \in [-1, 1], \text{ fix } n = 3$$



Grid search of (β_1, β_2) on Reddi et al. counterexample

Standard assumptions in the existing works

Commonly used assumptions in literature: A1 – A4

A1 (L-smooth): assume $\nabla \ell_i(\theta)$ are L-Lipschitz continuous

A2 (bounded Variance): $\frac{1}{n} \sum_{i=1}^n \|\nabla \ell_i(\theta) - \frac{1}{n} \sum_{i=1}^n \nabla \ell_i(\theta)\|_2^2 \leq D_0$

A3 (bounded gradient): $\|\nabla \ell(\theta)\| < C$

A4 (bounded 2nd order momentum): $v_k \in [C_l, C_u]$

[Reddi et al.18, Luo et al.19, Chen et al.18, Zou et al. 19, Chen et al. 22...]; proposed and studied variants of Adam
[Defossez et al. 2022] studied Adam: but A3 and A4 **avoid divergence a priori, reduced to AdaBound**

Standard assumptions in the existing works

We **removed A3 & A4** and **relaxed A2**

A1 (L-smooth): assume $\nabla \ell_t(\theta)$ are L-Lipschitz continuous

A2 (Affine Variance): $\frac{1}{n} \sum_{i=1}^n \|\nabla \ell_t(\theta) - \frac{1}{n} \sum_{i=1}^n \nabla \ell_i(\theta)\|_2^2 \leq D_1 \|\nabla \ell(\theta)\|_2^2 + D_0$

~~**A3 (bounded gradient):** $\|\nabla \ell(\theta)\| \leq G$~~

~~**A4 (bounded 2nd order momentum):** $v_k \in [C_{\ell}, C_u]$~~

[Reddi et al.18, Luo et al.19, Chen et al.18, Zou et al. 19, Chen et al. 22...]; proposed and studied variants of Adam
[Defossez et al. 2022] studied Adam: but A3 and A4 **avoid divergence a priori, reduced to AdaBound**

- A1 and A2 do not exclude Reddi's counterexample a priori
- A2 is weaker than bounded variance condition ($D_1 = 0$), mildest variance condition so far
- [Shi et al. 2021]: studied random-shuffling RMSProp (momentum-less Adam)
- We consider more general cases: both **with-replacement** & **random-shuffling Adam**; **(β_1, β_2) interaction**

Key findings: Concentration effect under large β_2

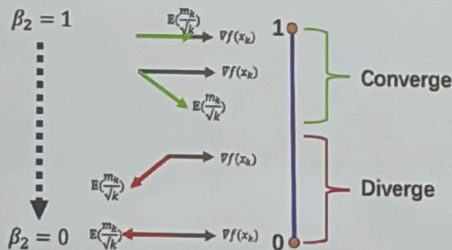
- Lemma 1: for any $0 < \delta \leq \frac{1}{4n}$, then with probability at least $1 - e^{-\frac{\delta}{1-\beta_2}}$, we have ($E_k := E(\cdot | \mathcal{F}_k)$)

$$\frac{C_{lower}}{\sqrt{E_k(v_k)}} \leq \frac{1}{\sqrt{v_k}} \leq \frac{C_{upper}}{\sqrt{E_k(v_k)}}$$

$$C_{lower} := 1 - (1 - \beta_2) \frac{4n}{(1 - 2n\delta)\beta_2^n}, \quad C_{upper} := \left(1 - (1 - \beta_2) \frac{8n}{(1 - 2n\delta)\beta_2^n} \right)^{-1/2}.$$

Remark: both $C_{lower}, C_{upper} \rightarrow 1$ as $\beta_2 \rightarrow 1$, and then $E_k \left(\frac{\nabla \ell_t(\theta)}{\sqrt{v_k}} \right) \approx \frac{E_k(\nabla \ell_t(\theta))}{\sqrt{E_k(v_k)}} = \frac{\nabla \ell(\theta)}{\sqrt{E_k(v_k)}}$, which forms a descent direction

$$\text{Adam: } x_{k+1} = x_k - \eta_k \frac{m_k}{\sqrt{v_k}}$$



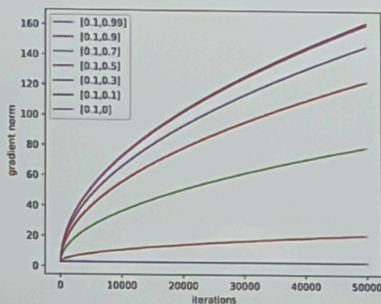
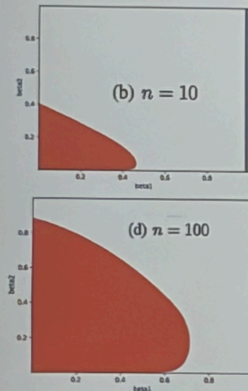
Proof Idea: track $1 / \sqrt{v_k}$ by a sequence of i.i.d. Bernoulli random variables + several decoupling tricks

Why Bernoulli? Randomness comes from n -sum sampling (Adaptive Learning Smooth) + EMA controls the perturbation $|\theta_k - \theta_{k-1}|$

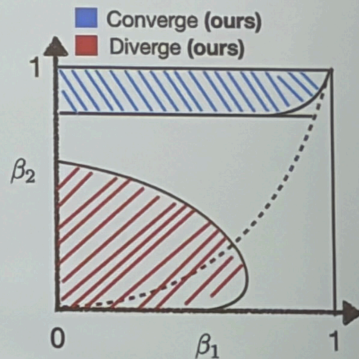
Phase Transition: Converge $\rightarrow$ Diverge

Theorem 1: when $\beta_2 \geq 1 - O\left(\frac{1-\beta_1^n}{n^{3.5}}\right)$, $\beta_1 < \sqrt{\beta_2} < 1$, Adam converges with rate $O\left(\frac{\log T}{\sqrt{T}}\right)$

Theorem 2: for $\forall n \geq 3$, $\exists \ell(\theta)$ in the same problem class as above, when (β_1, β_2) lies in the red region (also dependent on n), sequence of $\{\theta_k\}$ and $\{\ell(\theta_k)\}$ of Adam diverges to ∞



(b) The effect of small β_2



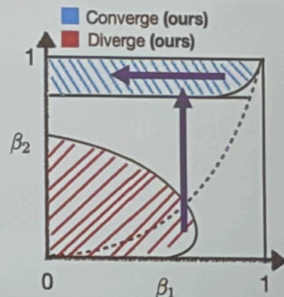
Phase Transition: Converge $\rightarrow$ Diverge

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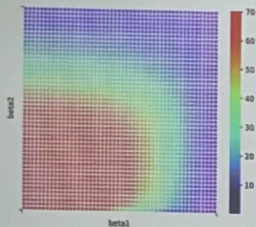
Suggestions for hyperparameter-tuning:

1. Increase β_2 **inversely** with batchsize to surpass β_2^*
2. Try different $\beta_1 < \sqrt{\beta_2}$ for better performance

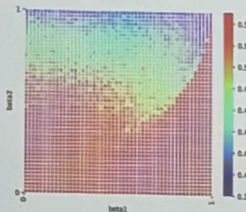


New Con/Di-vergence Region Matches Practice Better

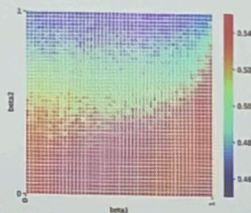
Fix n :



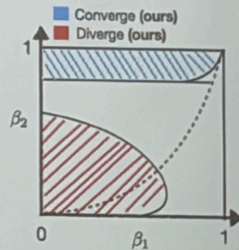
(a) Function 2



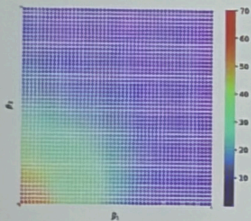
(b) MNIST



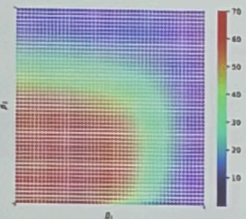
(c) CIFAR-10



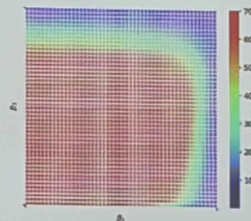
Change n :



(a) $n = 5$



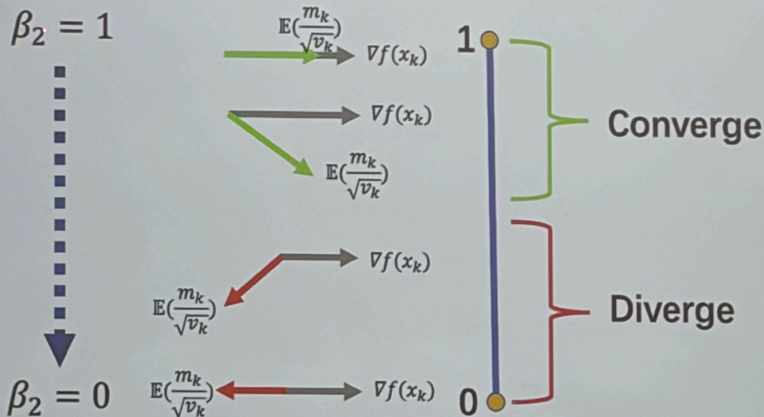
(b) $n = 10$



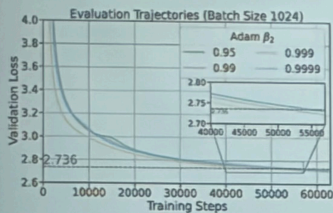
(d) $n = 20$

Intuition: β_2 primarily determines update direction

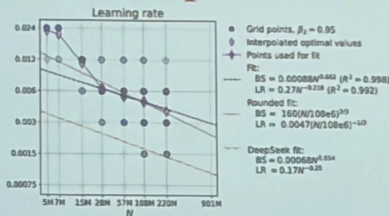
$$\text{Adam: } x_{k+1} = x_k - \eta_k \frac{m_k}{\sqrt{v_k}}$$



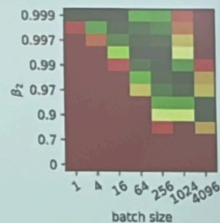
Feedback from LLM developers



(b) β_2 Ablation
[Figure from SOAP team]



G.3 The necessity of tuning β_2
[Figure from Yarn Carmon team]



[Figure from Marek et al. 25]

Recent findings by Harvard & Amazon:

“larger β_2 (increase from 0.95 to 0.99, 0.999, or 0.9995) substantially improves small batch size training... aligning with findings in [Zhang et al., 2022]”

Recent findings by Yair Carmon and coauthors (Intern of Ilya Sutskever):

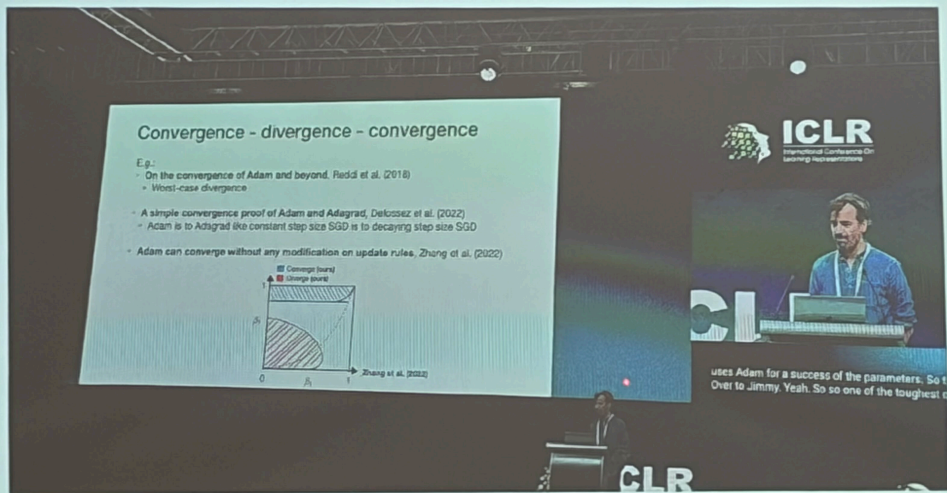
“...we find that tuning (enlarging) the β_2 (from 0.95 to 0.99 and 0.999) is essential at lower batch sizes... This matches the theoretical work by [Zhang et al. 2022] that showed that when the batch size is small, higher values of β_2 are crucial”

Recent findings by Max Planck Institute: “Zhang et al. [2022] shows that higher β_2 substantially improve small-batch training, and Marek et al. [2025] highlights the importance of scaling β_2 in this regime.”

H. Zhang, et al. HOW DOES CRITICAL BATCH SIZE SCALE IN PRETRAINING?, ICLR 2025
Srećković, et al. Is your batch size the problem? revisiting the adam-sgd gap in language modeling, preprint
Poriam, T et al., Resolving discrepancies in compute-optimal scaling of language models, NeurIPS 2024

Acknowledgements from the Authors of Adam

- Photo shot at **ICLR 2025 Test of Time Speech** by Dr. Durk Kingma and Prof. Jimmy Ba
- **Prof. Jimmy Ba**: acknowledged our result as the convergence guarantee; explained how to choose β_1, β_2 using our theory



References for this part

Adam Can Converge Without Any Modification On Update Rules

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Adam Converges Without Any Modification On Update Rules

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Conference version:

- Appeared online in Aug 2022
- NeurIPS 2022 (**Spotlight**)
- **Proof: 40+ pages**

Extended journal version:

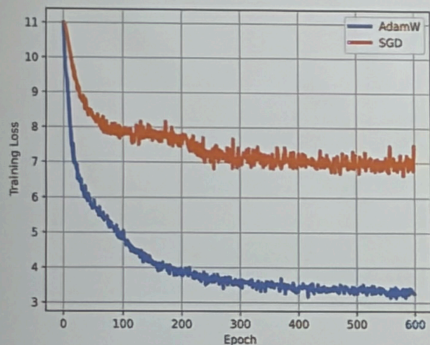
- Appeared online in March 2026
 - Under review at Mathematical Programming
- ### New contents:
- A simplified proof under random shuffling (**20+ pages**)
 - A new proof under with-replacement sampling (**20+ pages**)

The divergence-convergence of Muon

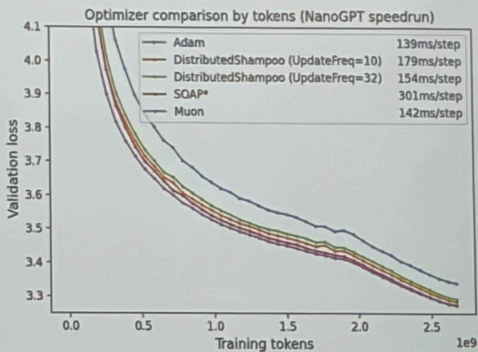
Direct implications to Muon:

- Muon also has **di-convergence phase transition**:
 - SignSGD = Adam with $\beta_1 = \beta_2 = 0$, diverges
 - SignSGD = Muon with $\beta = 0$ and $m = n = 1$, diverges
- To avoid divergence: **use large β** for Muon, depends on batch size
- Haven't seen empirical studies on sweeping β of Muon (usually fix $\beta = 0.95$)
- Haven't seen any of such theory for now

SGD < AdamW < Muon for LLMs



GPT2-125M: SGD < AdamW



GPT2-125M: AdamW < Muon

SGD < AdamW < Muon, but why? Let us start with why SGD < Adam

Let us start with SGD v.s. Adam

Major difference of Adam with SGD: its **diagonal preconditioner**

$$w \leftarrow w - \eta D \circ m$$

-- SGD: $D_{SGD} = I$

-- Adam: $D_{Adam} = \text{Diag}(\frac{1}{\sqrt{v_1}}, \frac{1}{\sqrt{v_2}}, \dots, \frac{1}{\sqrt{v_d}})$

- In numerical linear algebra: diagonal preconditioner is NOT always effective
- closely related to **Hessian structure**: dense, sparse, diagonal, diagonal-dominant... [1,2]

[1] Worst-case Complexity of Cyclic Coordinate Descent, Sun and Ye, 2017, Mathematical Programming

[2] Optimal Diagonal Preconditioning, Qu, Gao, Hinder, Ye, and Zhou, 2022, Operation Research

Why SGD < Adam < Muon? A preconditioner perspective

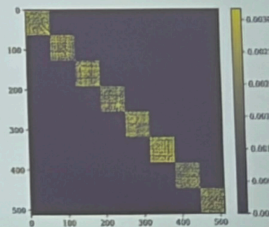
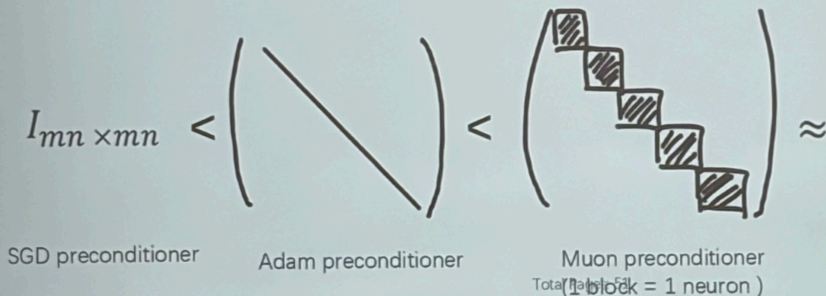
- Matrix form of Muon: $W \in \mathbb{R}^{m \times n}$

$$W_{k+1} = W_k - \eta_k (M_k M_k^T)^{\frac{1}{2}} M_k = W_k - \eta_k I_m M_k (M_k^T M_k)^{\frac{1}{2}}$$

- Flattened vector form of Muon:

$$\text{vec}(W_{k+1}) = \text{vec}(W_k) - \eta_k \left(I_m \otimes (M_k^T M_k) \right)^{\frac{1}{2}} \text{vec}(M_k)$$

we used the proposition of Kronecker product: $A \otimes B \text{vec}(C) = ACB$



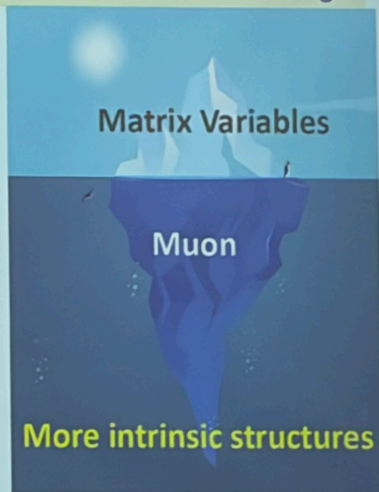
Hessian structure
(1 block = 1 neuron)

For multi-layer models:

- Source of special Hessian:
consecutive multiplication of **large** and **well-shaped** matrix variables + training

$$y = W_3 W_2 W_1 x$$

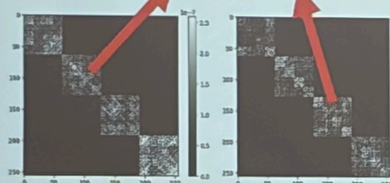
Rigorous theory: ongoing...



Hessian of Transformers?

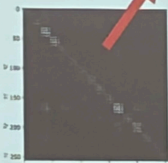
#blocks = #attention heads

Less clear (treat as #output neurons)

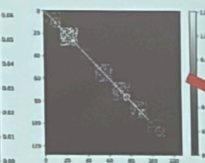


(a) query (4 heads)

(b) key (4 heads)

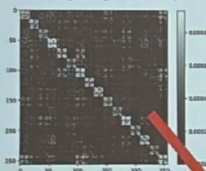


(c) value (4 heads)

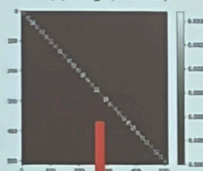


Embedding layer (8 tokens)

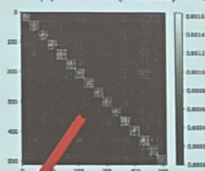
#blocks
= #tokens



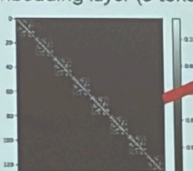
(d) attn.proj (16 neurons)



(e) mlp.fc1 (32 neurons)



(f) mlp.proj (16 neurons)



Output layer (8 tokens)

#blocks = #output neurons

Contents

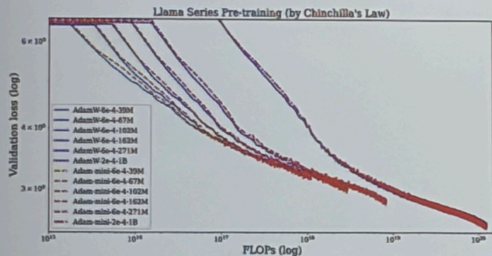
- Part I: Does Adam Diverge or Converge?
- Part II: Why $\text{SGD} < \text{Adam} < \text{Muon}$ on Transformers?
- Part III: How to improve Adam and Muon?
- Part IV: Conclusions

Adam-mini: use neuron-wise lr instead of coordinate-wise lr

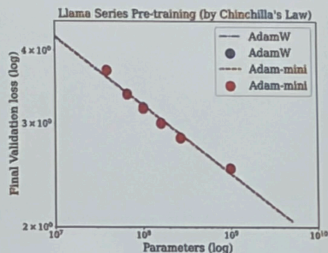
- Step 1: partition the gradient g into B sub-vectors according to the dense Hessian sub-block $g_b, b = [B]$
- Step 2: for each g_b , calculate:

$$v_b = (1 - \beta_2) * \text{mean}(g_b \odot g_b) + \beta_2 * v_b, \quad b = 1, \dots, B.$$

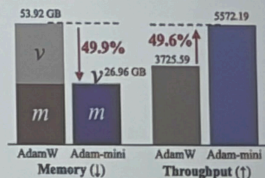
This removes 99.9% v in Adam, but the final 0.1% is crucial for performance



(a) Scaling laws in terms of compute



(b) Scaling laws in terms of parameters



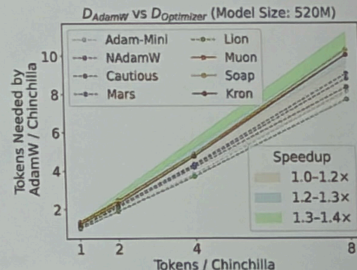
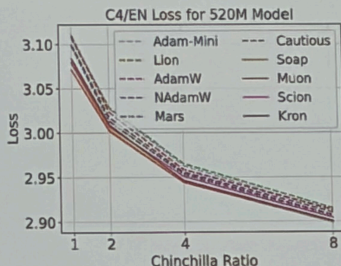
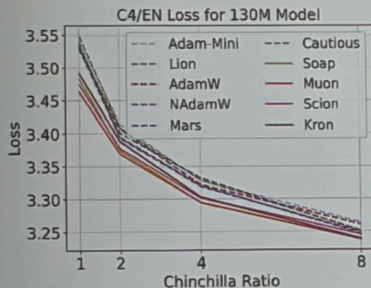
Chinchilla Scaling laws of Adam-mini: same performance as AdamW, but with **50%** less memory

Adam-mini (matrix equivalent form): Row-wise normalization

- Adam-mini for non-attention blocks:
Given a hidden-layer gradient matrix $G \in \mathbb{R}^{m \times n}$;
 - $M_k = (1 - \beta_1)G + \beta_1 M_{k-1} \in \mathbb{R}^{m \times n}$
 - $vmean_k = (1 - \beta_2) \text{rowmean}(G \circ G) + \beta_2 vmean_{k-1} \in \mathbb{R}^m$
 - $W_{k+1} = W_k - \eta_k M_k / vmean_k$
- Recently, there is a surge of **new row-wise normalized methods**:
(RMNP, Mano, Aurora, Nora...)
 - Some are motivated by Adam-mini, some are independently designed
 - Experiments show that these methods **can even match Muon**.
 - Curious how they would perform in larger scales

Optimizer Benchmark by Stanford

- 130/ 300/ 520M model with 1x / 2x / 4x / 8x Chinchilla token amount
- Full hyperparameter sweep for all optimizers
- On 520M model, Adam-mini is **slightly faster than** AdamW



"Adam-mini closely tracks the performance of AdamW ... and sometimes even performs better than AdamW"

Wen, Hall, Ma, and Liang, (2025). Fantastic pretraining optimizers and where to find them. arXiv preprint arXiv:2509.02046.

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We have discussed Adam ... How to improve Muon?

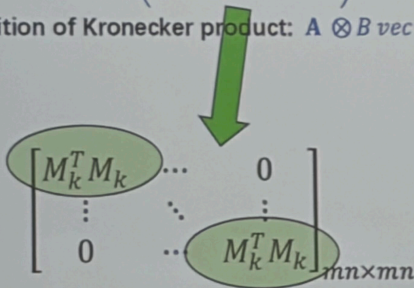
- Matrix form of Muon: $W \in R^{m \times n}$

$$W_{k+1} = W_k - \eta_k (M_k M_k^T)^{-\frac{1}{2}} M_k = W_k - \eta_k I_m M_k (M_k^T M_k)^{-\frac{1}{2}}$$

- Flattened vector form of Muon:

$$\text{vec}(W_{k+1}) = \text{vec}(W_k) - \eta_k \left(I_m \otimes (M_k^T M_k)^{-\frac{1}{2}} \right) \text{vec}(M_k)$$

We used the proposition of Kronecker product: $A \otimes B \text{vec}(C) = ACB$



$$\begin{bmatrix} M_k^T M_k & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & M_k^T M_k \end{bmatrix}_{mn \times mn}$$

Key observation:

Muon use the **same** preconditioner across neurons ...

But neurons in Transformers are **heterogeneous...** [1, 2]

How to improve Muon?

[1] Why Transformers Need Adam: A Hessian Perspective, NeurIPS 2024

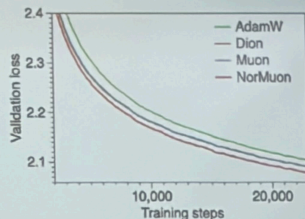
[2] Technical report of Aurora optimizer May 2026

Solution: Adam-mini + Muon

- **NorMuon**: add **neuronwise lr** to Muon
- by Microsoft and GaTech, including an author of LoRA

Algorithm 1 NorMuon

- 1: **Input**: Initial weights $W_0 \in \mathbb{R}^{m \times n}$, loss L , learning rate η , momentum parameters (β_1, β_2) , perturbation parameter ε , weight decay λ .
- 2: Initialize $M_0 \in \mathbb{R}^{m \times n} \leftarrow 0$, $v_0 \in \mathbb{R}^m \leftarrow 0$
- 3: **for** $t = 1, 2, \dots$ **do**
- 4: $G_t \leftarrow \nabla_W L(W_t)$
- 5: $M_t \leftarrow \beta_1 M_{t-1} + (1 - \beta_1) G_t$
- 6: $O_t \leftarrow \text{NSS}(M_t)$
- 7: $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) \text{mean}_{\text{cols}}(O_t \odot O_t)$
- 8: $V_t \leftarrow \text{ExpandRows}(v_t)$
- 9: $\tilde{O}_t \leftarrow O_t \odot (\sqrt{V_t} + \varepsilon)$
- 10: $\hat{\eta} = 0.2\eta \sqrt{mn} / \|\tilde{O}_t\|_F$
- 11: $W_{t+1} \leftarrow W_t - \hat{\eta} \lambda W_t - \hat{\eta} \tilde{O}_t$
- 12: **end for**



(a) Pretraining results of 1.1B model.

NorMuon: Making Muon more efficient and scalable

Zichong Li¹, Liming Liu^{1†}, Chen Liang², Weizhu Chen², Tuo Zhao¹
¹Georgia Tech ²Microsoft

This observation motivates our key insight: while Muon's orthogonalization effectively reduces the condition number of updates, the remaining high variance in neuron norms still creates an imbalanced learning dynamic, potentially leading to inefficient parameter usage. Drawing inspiration from Adam-mini's success [Zhang et al. \(2025\)](#) with per-neuron adaptive learning rates, we propose to incorporate second-order momentum to normalize these disparate scales and ensure more balanced parameter updates. Our method, **NorMuon**, augments Muon's orthogonalization with neuron-wise adaptive learning rates computed from accumulated second-order statistics.